Making a CPM of Collective Migration

Report on Assignment 1C

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1 Introduction

Collective cell migration concerns the movement of a group of cells in response to environmental or chemical cues, and is an emergent property of many interacting individual cells. In this phenomenon, cells in the back of a group can push the front cells to move quicker than they otherwise would have. Cells in the back can place pixels inside cells that are in front of them, thereby putting the front cells 'under volume'. This results in the front cells placing new pixels in front of them, resulting in faster cell movement. We can explore how individual cells interacting with their environment give rise to such collective behavior, by using the Cellular Potts Model (CPM).

Obstacles represent some environmental constraints that affect collective cell migration by guiding the movement of cells or disrupting it. By varying the number of obstacles, changes in the collective cell migration can be observed. Investigating these effects helps reveal the underlying mechanisms that guide cell and group behaviors. This experiment aims to explore how obstacles change the collective migration behavior of cells by observing simulations with 0, 9, 25, and 49 obstacles.

The goal of this project is to use CPM to investigate the impact of obstacles on collective cell migration.

2 Methodology

The Artistoo framework was used to implement the two-dimensional CPM simulations. The model was defined on a square grid of 200x200 pixels with wrapped dimensions. In addition to the background, there were two types of cells in the simulations: moving cells and obstacles. Obstacles were implemented by creating a new type of cell with different parameters in order to distinguish the two types of cells and to give them desired properties. All parameters used in this project are displayed in Table 1.

Obstacles were initialized with a target volume (V) of 100 pixels and volume constraint strength (λ_V) of 500 to ensure they are static and do not deform. Their perimeter (P) was set to 50 with a strong perimeter constraint strength (λ_P) of 200, keeping their shape strongly constrained. To further prevent any movement, their activity ($\lambda_{\text{act}}, \text{Max}_{\text{act}}$) was set to 0. Additionally, the adhesion between obstacles and moving cells ($J_{\text{obstacle-cell}}$) was set to 1000 to ensure moving cells do not stick to obstacles or penetrate them. Lastly, obstacles were initialized with obstacle padding. This padding is the number of pixels between the border and obstacles. All obstacles within this bounding box are equally spaced in a grid-like pattern.

On the other hand, moving cells had completely different parameters that would allow them to move and explore their environment, reproducing collective cell migration. These parameters were directly taken from the self-study exercise 1.3, as they are known to reproduce collective migration. A total of 80 moving cells were placed on the grid in each simulation. 80 was chosen because it does not clutter the screen and it allows for the observation of collective cell migration. Each moving cell was initialized with V of 200 pixels, making them twice the size of the obstacles, and λ_V of 50 to allow them moderate compressibility. Their P was set to 180 and λ_P to 2 to allow them to take on irregular shapes during their migration. Their movement, and the active migration, was controlled using activity constraints with λ_{act} of 200 and Max_{act} 20. These parameters allowed the reproduction of collective migration.

Table 1: CPM parameters for moving cells and obstacles.

Parameter	Moving Cells	Obstacles
Temperature (T)	20	20
Adhesion cell–matrix ($J_{\text{obstacle-matrix}}$)	20	20
Adhesion cell–cell ($J_{\text{cell-cell}}$)	0	0
Adhesion obstacle–cell ($J_{\text{obstacle-cell}}$)	1000	1000
Target volume (V)	200	100
Volume constraint (λ_V)	50	500
Target perimeter (P)	180	50
Perimeter constraint (λ_P)	2	200
Activity strength (λ_{act})	200	0
Activity memory (Max_{act})	$20 \rightarrow 60$	0
Number of cells	80	0, 9, 25, 49
Obstacle padding	–	10

Experimental design

Eight experiments were performed in total, organized into two subexperiments of four. The variables changed was the number of obstacles (0, 9, 25, and 49), and the value of Max_{act}. The baseline experiments used $\text{Max}_{\text{act}} = 20$, and the adjusted experiments used $\text{Max}_{\text{act}} = 60$. All other parameters were identical across all eight simulations (Table 1). The system temperature was fixed at $T = 20$ for all runs, controlling the stochasticity of the model. Each simulation was initialized by placing the obstacles first, and then randomly seeding moving cells. Each simulation ran for 2000 Monte Carlo steps, which took about 10 minutes in wall-clock time at 1 frame per second, and the observations and results are reported in the Results section.

To get quantitative data, a Python script was used to calculate the average speed and average distance traversed, as well as find the fastest and slowest cells, along with the standard deviation in speeds. It also calculates the Vicsek order parameter for each timestep and takes the mean, giving us data on how stable the alignment of moving cells is. This allows us to analyze the cell movements for both environments with and without obstacles in a quantitative, objective way, and base our qualitative analysis on facts.

3 Results

Qualitative observations

In the baseline experiments(Figure 1), cells initialized randomly and, after several Monte Carlo steps the clustered into groups. The groups migrated collectively in the sense that cells moved in coordination and maintained contact, but the net displacement of the group as a whole was small. Cells tended to explore without a sustained preferred direction, resulting in low-persistence collective motion rather than directed migration. This is likely caused by $\text{Max}_{\text{act}} = 20$, which prevented the group from aligning persistently.

The adjusted experiments (Figure 2) produced noticably different behavior in the obstacle-free condition. With $\text{Max}_{\text{act}} = 60$, we see large groups that not only maintained cohesion but also migrated persistently and rapidly in a single shared direction, covering substantially greater distances than in the baseline experiments. This represents a qualitative shift from low-persistence collective exploration to directed collective migration.

Introducing obstacles in the adjusted experiments progressively disrupted this collective behavior. When a cell encounters an obstacle, it easily avoids it, and the underlying cell behavior does not change. At 9 obstacles, the effect was negligible(Figures 2d, h, l). However at 25 obstacles, group sizes decreased and movement became more spatially constrained, with cells forced to migrate in one of 4 directions. (Figures 2c, g, k). At 49 obstacles, cells frequently moved alone or in very small clusters that changed direction often due to collisions with obstacles.

Quantative observations

The quantitative results for the baseline experiments are summarized in Table 2. Despite the low-persistence character of migration at $\mathrm{Max}_{\mathrm{act}} = 20$, a clear decrease in both average speed and average distance traveled is observed as obstacle density increases. From 0.0347 pixels/step and 82.95 pixels at 0 obstacles down to 0.0300 pixels/step and 71.76 pixels at 49 obstacles, which is a reduction of approximately 14% in speed and 13% in distance. Notice that the fastest individual cell at 49 obstacles (0.0814 pixels/step) substantially exceeds that at 0 obstacles (0.0614 pixels/step), possibly because isolated cells are channeled by obstacles into brief stretches of unimpeded motion. The slowest cell speed drops markedly in the presence of obstacles, consistent with occasional blockage.

Table 2: Speed and distance results (pixels/step) of CPM with baseline parameters.

Amount of Obstacles	**0**	**9**	**25**	**49**
Avg speed across all cells	0.0347	0.0340	0.0324	0.0300
Std dev of cell speeds	0.0075	0.0094	0.0070	0.0107
Fastest cell avg speed	0.0614	0.0614	0.0597	0.0814
Slowest cell avg speed	0.0239	0.0044	0.0198	0.0026
Avg distance across all cells	82.95	81.36	77.47	71.76

Table 3 shows results for the adjusted experiments. Compared to the baseline the average speeds are approximately 3.4× higher (0.1168 vs. 0.0347 pixels/step at 0 obstacles), and average distances are more than three times greater (279.12 vs. 82.95 pixels). This increase directly reflects the directed, persistent character of collective migration at $\mathrm{Max}_{\mathrm{act}} = 60$, because groups sustain a shared direction over many Monte Carlo steps. Despite these higher absolute values, the same trend can be observed. We see that the average speed and distance decrease with obstacle density, from 0.1168 pixels/step and 279.12 pixels at 0 obstacles to 0.1008 pixels/step and 240.95 pixels at 49 obstacles. This is a reduction of roughly 14% in both metrics, nearly identical in proportion to the baseline experiments.

Table 3: Speed distance results (pixels/step) of CPM with adjusted parameters.

Amount of Obstacles	**0**	**9**	**25**	**49**
Avg speed across all cells	0.1168	0.1100	0.1094	0.1008
Std dev of cell speeds	0.0070	0.0116	0.0095	0.0131
Fastest cell avg speed	0.1366	0.1333	0.1319	0.1316
Slowest cell avg speed	0.1005	0.0616	0.0917	0.0677
Avg distance across all cells	279.12	262.96	261.54	240.95

4 Discussion

This experiment explores how different obstacle configurations impact collective cell migration and highlights two distinct findings. First, $\mathrm{Max}_{\mathrm{act}}$ fundamentally determines the mode of collective migration; at $\mathrm{Max}_{\mathrm{act}} = 20$, cells form cohesive groups but wander without a persistent net direction, whereas at $\mathrm{Max}_{\mathrm{act}} = 60$, the cells sustain their direction of motion, resulting in large groups that migrate persistently and rapidly in a shared direction, covering more than three times the distance. Second, increasing obstacle density consistently disrupts collective migration in both series. At low densities (9 obstacles), the effect is minor. At intermediate densities (25), groups fragment and become channeled between obstacles. At high densities (49), sustained collective motion is largely abolished and cells migrate individually or in small clusters. These effects are reflected in the quantitative data, which show a decrease of approximately 14% in both average speed and distance as obstacle density increases from 0 to 49, consistent across both parameter series

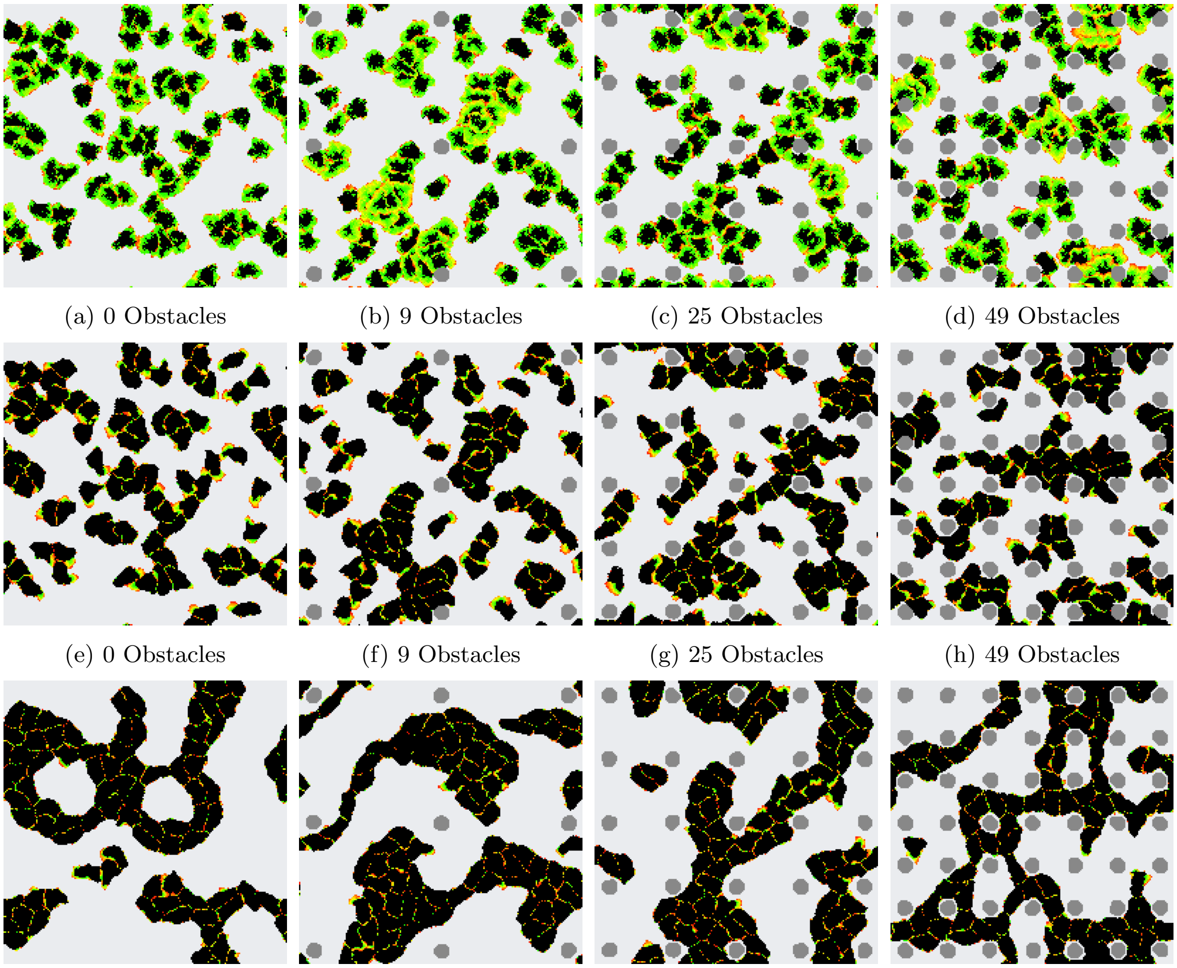
Several limitations should be acknowledged. All obstacles were arranged in a regular grid, which introduces an artificial directional bias and does not reflect the irregular geometries found in biological environments. Only four obstacle densities were tested and most parameter values were taken from the self-study exercise 1.3, so statistical robustness cannot be assessed. It is recommended that future studies explore different parameter configurations, irregular obstacle placement, and dynamic obstacles.

5 Conclusion

This study investigated the effect of obstacles on collective cell migration using a two-dimensional Cellular Potts Model implemented in Artistoo, across eight simulations spanning two parameter series and four obstacle densities. Without obstacles, cells formed large groups that coherently moved together in the same direction. Increasing the obstacle density disrupted collective migration. Low obstacle densities only had minor effects, however, as the density increases, group sizes and cell movement become more restricted. This is also supported by the quantitative results which show consistent decrease in average speed and in distance traveled as the number of obstacles grows.

Overall, this work demonstrates that obstacles play a critical role in collective cell behavior.

Appendices

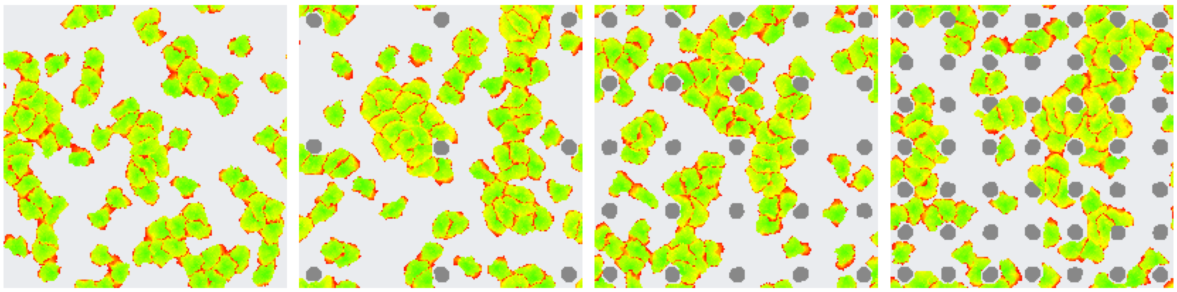


(a) 0 Obstacles (b) 9 Obstacles (c) 25 Obstacles (d) 49 Obstacles

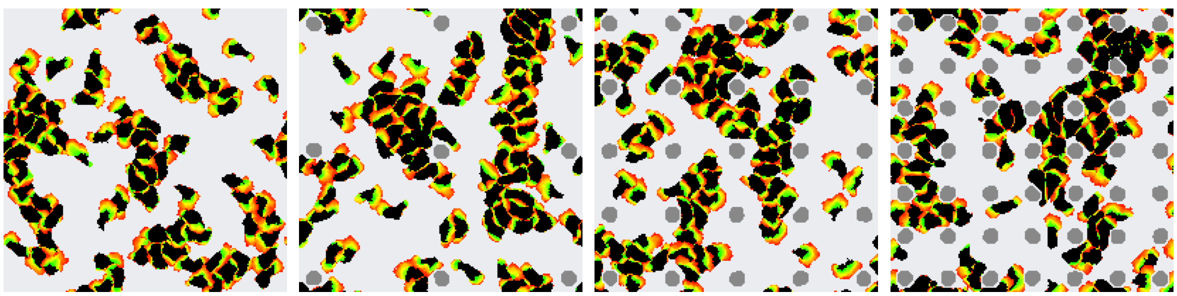
(e) 0 Obstacles (f) 9 Obstacles (g) 25 Obstacles (h) 49 Obstacles

(i) 0 Obstacles (j) 9 Obstacles (k) 25 Obstacles (l) 49 Obstacles

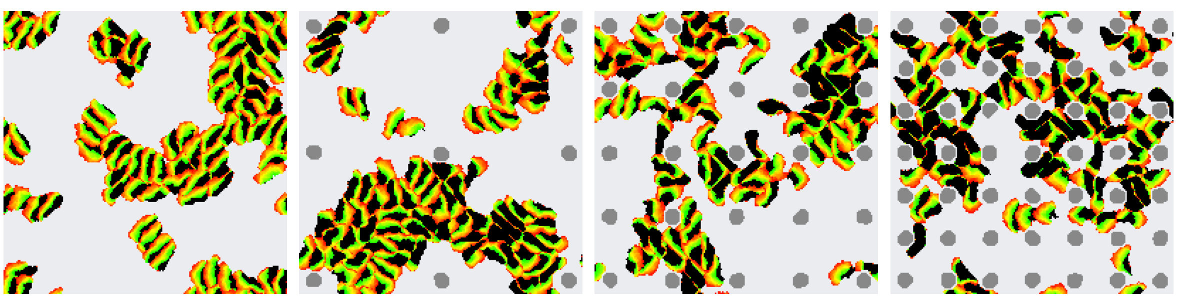
Figure 1: 80 Cells in an environment with different amount of obstacles with baseline parameters.



(a) 0 Obstacles (b) 9 Obstacles (c) 25 Obstacles (d) 49 Obstacles



(e) 0 Obstacles (f) 9 Obstacles (g) 25 Obstacles (h) 49 Obstacles



(i) 0 Obstacles (j) 9 Obstacles (k) 25 Obstacles (l) 49 Obstacles

Figure 2: 80 Cells in an environment with different amount of obstacles with adjusted parameters.